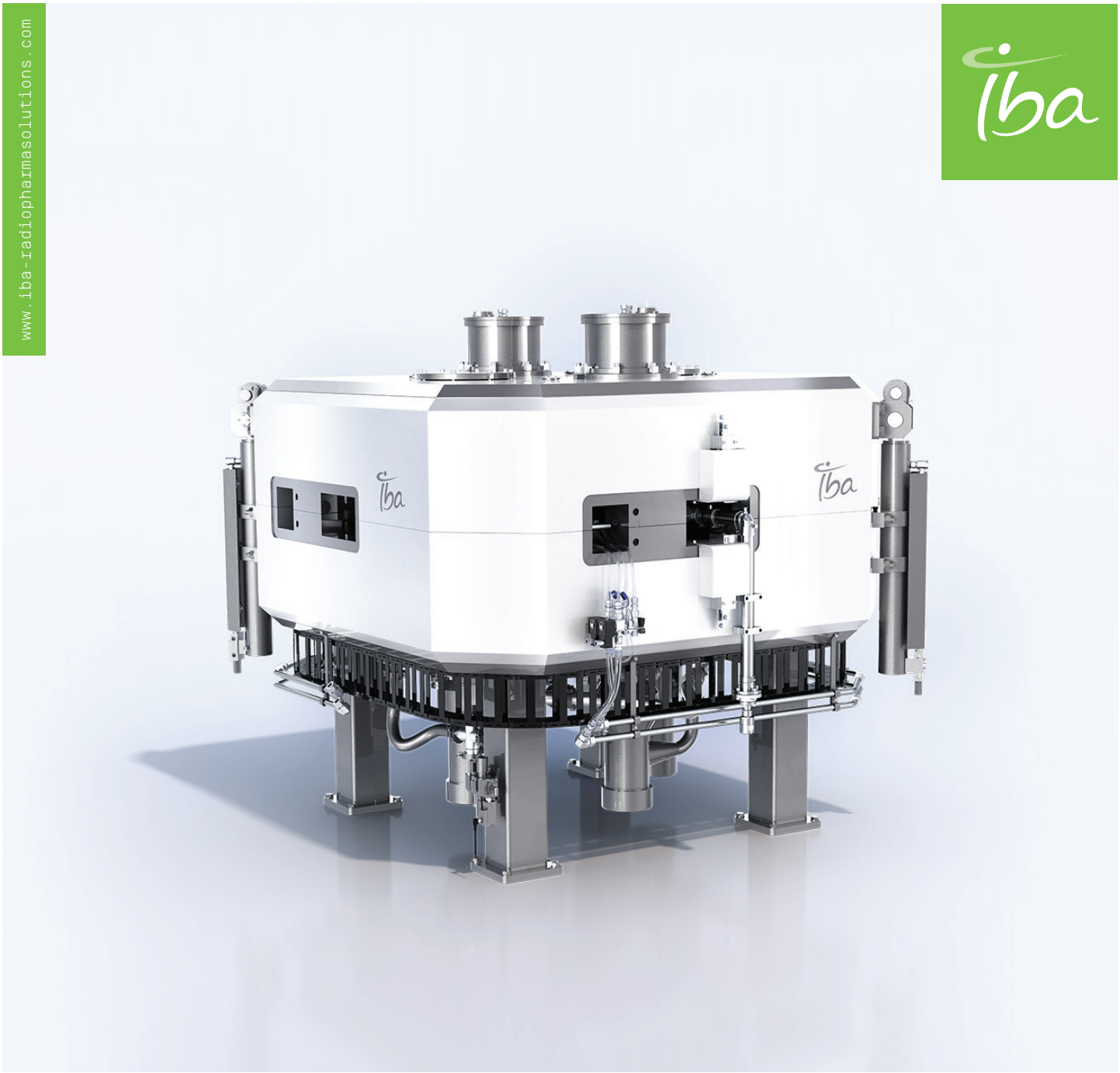
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CYCLONE®KIUBE

MAX POTENTIAL, MAX CAPACITY

PROTECT +
ENHANCE +
SAVE LIVES

RADIO
PHARMA
SOLUTIONS

30 years of experience concentrated in one innovation



A STORY OF CYCLOTRON INNOVATIONS

1985
DEEP VALLEY CYCLOTRON
5 X MORE OUTPUT, 3 X LESS ENERGY
EP022786

1990
VECTIO® BEAM
TRANSPORT LINE
FOR R&D

2000
NIOBIUM NIRTA® FLUOR TARGET
FOR BETTER ISOTOPE QUALITY
EP1716576

2002
NIRTA® SOLID TARGET
TO GIVE ACCESS TO NEW ISOTOPES
EP1570493

2003
DUAL BEAM REGULATION
FOR DOUBLED PRODUCTION
EP1566082

2005
HIGH CURRENT MACHINE 150µA
FOR INCREASED PRODUCTION
CAPACITY

2006
TARGET SHIELDS FOR
EASY DECOMMISIONING
EP200.199

2008
TWIN PROTON SOURCES
TO MAXIMIZE UPTIME
EP2196073

2011
NIRTA® FLUOR CONICAL TARGET
FOR REDUCED ENRICHED WATER USE
WO 2012/055970

2012
ZEPHIROS® CYCLOTRON
CONTROL SYSTEM
FOR EASE OF USE
EP2581914

2015
GA-68 PRODUCTION
USING A LIQUID TARGET
EP15170854

2016
CYCLONE® KIUBE
EP16169489, EP16169490
EP16169494, EP16169497
EP16171282

Max potential, Max capacity

30 YEARS OF INNOVATION AND EXPERTISE CONCENTRATED IN THE CYCLONE® KIUBE

The first world reference

In 1986, IBA launched the first compact cyclotron for industrial production of radioisotopes. This revolutionary new design increased the production efficiency by a factor 15 and created a world reference.

30 years on, IBA once again makes history with the launch of its new 18 MeV cyclotron. Cyclone® KIUBE is more compact and the most powerful PET cyclotron. It has been developed with three keywords in mind: **Reliability, High Performance and Flexibility.** These keywords were defined in tangent with our Users**.**

Expertise benefiting to a large installed base

With IBA as partner, you have the assurance to remain at the forefront of innovation and to benefit of every novel feature for tracer production.

More than 250 cyclotrons installed over 5 continents demonstrate this expertise and customer satisfaction.

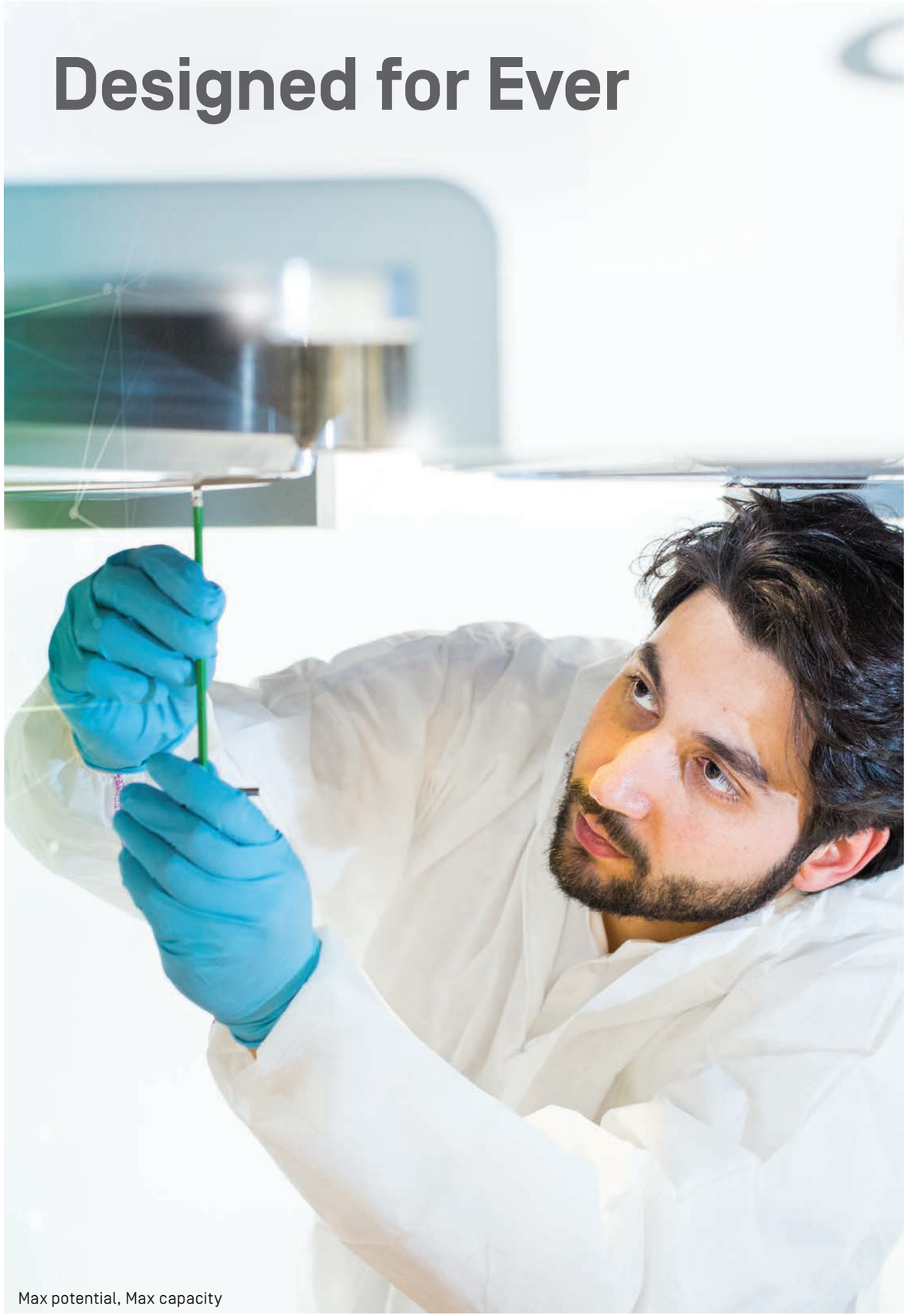
Support over the lifetime of your system

IBA provides continuous support over the lifetime of your project, expanding your skills and applications, boosting your uptime and maximizing the return on your investment with all the latest innovations, services and training.
In addition the system is built so that future upgrades can easily be implemented to keep your cyclotron state-of-the-art.

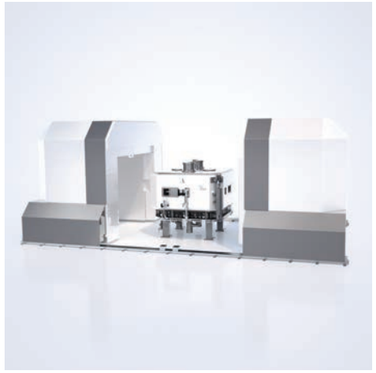
150
IBA patents on particle accelerators

30
Years of expertise

Designed for Ever



Max potential, Max capacity



CYCLONE® KIUBE WILL EVOLVE WITH YOU, FOR YOU

Upgradable like no other

Cyclone® KIUBE is upgradable like no other cyclotron, so you can increase your capacity step-by-step; from 100 µA to 150µA, 180 µA and even up to 300µA on target.

With a lower initial investment, you can start with the Cyclone® KIUBE 100 while maintaining the ability to increase your PET Center's capacity over time.

Infinite flexibility

With eight exits ports, Cyclone® KIUBE is the most flexible system, producing the widest range of PET radioisotopes. Efficiency with an edge.

A full range of Nirta® targets are available giving you access to **^{18}F, ^{13}N, ^{15}O, ^{11}C** (CO_2 & CH_4), **$^{18}F_2$, ^{68}Ga, ...**

In addition, IBA provides Nirta® Solid target technology for the production of radioisotopes such as **^{64}Cu, ^{89}Zr, ^{124}I, ^{99m}Tc...**

This is our typical production schedule thanks to the IBA 8-exit ports cyclotron:

Time	Radio-nuclide	Radio pharmaceutical
4.30-6.00	^{18}F	FDG
6.30-8.00	^{18}F	F-other
8.15-8.45	^{11}C	CHOLINE
10.00-12.00	^{18}F	MK9470
12.30-13.00	^{11}C	Raclopride
13.30-13.40	^{13}N	NH3
14.00-16.00	^{15}O	10*H_2O
16.30-17.00	^{11}C	PDE10A

Prof. Dr GUY BORMANS
Head of radiopharmacy
Katoliek Universiteit Leuven, Belgium

CYCLONE® KIUBE WITH SELF-SHIELDING OPTION*



10Ci 370 GBq 100 µA → **16Ci** 592 GBq 150 µA

UPGRADABLE

CYCLONE® KIUBE VAULTED*



10Ci 370 GBq 100 µA → **16Ci** 592 GBq 150 µA → **20Ci** 740 GBq 180 µA → **30Ci** 1110 GBq 300 µA

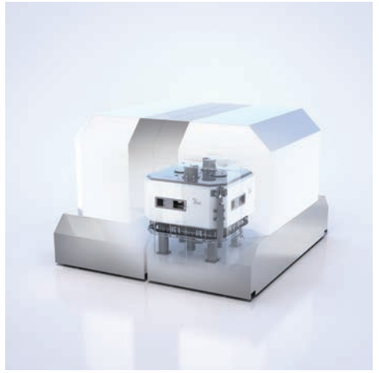
UPGRADABLE

*Production in Curies of $^{18}F^-$ in 2 hours run

Designed to Deliver



Max potential, Max capacity



CYCLONE® KIUBE DELIVERS OUTSTANDING PERFORMANCES

The most cost-effective

It has been demonstrated that 18 MeV is the optimal proton energy for the highest production yield of most of PET radioisotopes. Your production capacity is optimized so as your revenue; your center footprint is kept to a reasonable size so as your investment. This means that 18 MeV is the most cost-effective solution for your isotope production.

The growing demand for radioisotopes means a greater need for efficiency. Cyclone® KIUBE saves enriched water and has the lowest power consumption per Curie produced, yet is the most powerful of the market.

The most performant cyclotron

Cyclone®KIUBE offers unmatched production capacity for an internal source PET cyclotron.
A 2-hours dual beam run could generate up to 30 Ci of ^{18}F-; hence, an incredible batch of FDG synthesized on your Synthera®+ modules.

Maximum reliability

Its unique twin-proton sources give maximum reliability, as switching to the second source during operation is simple, fast and completely automated, translating into more than 99% source uptime.
Moreover, the production is optimized in real time thanks to the automatic tuning of the ion source, stripper and magnetic field.

300 (µA)

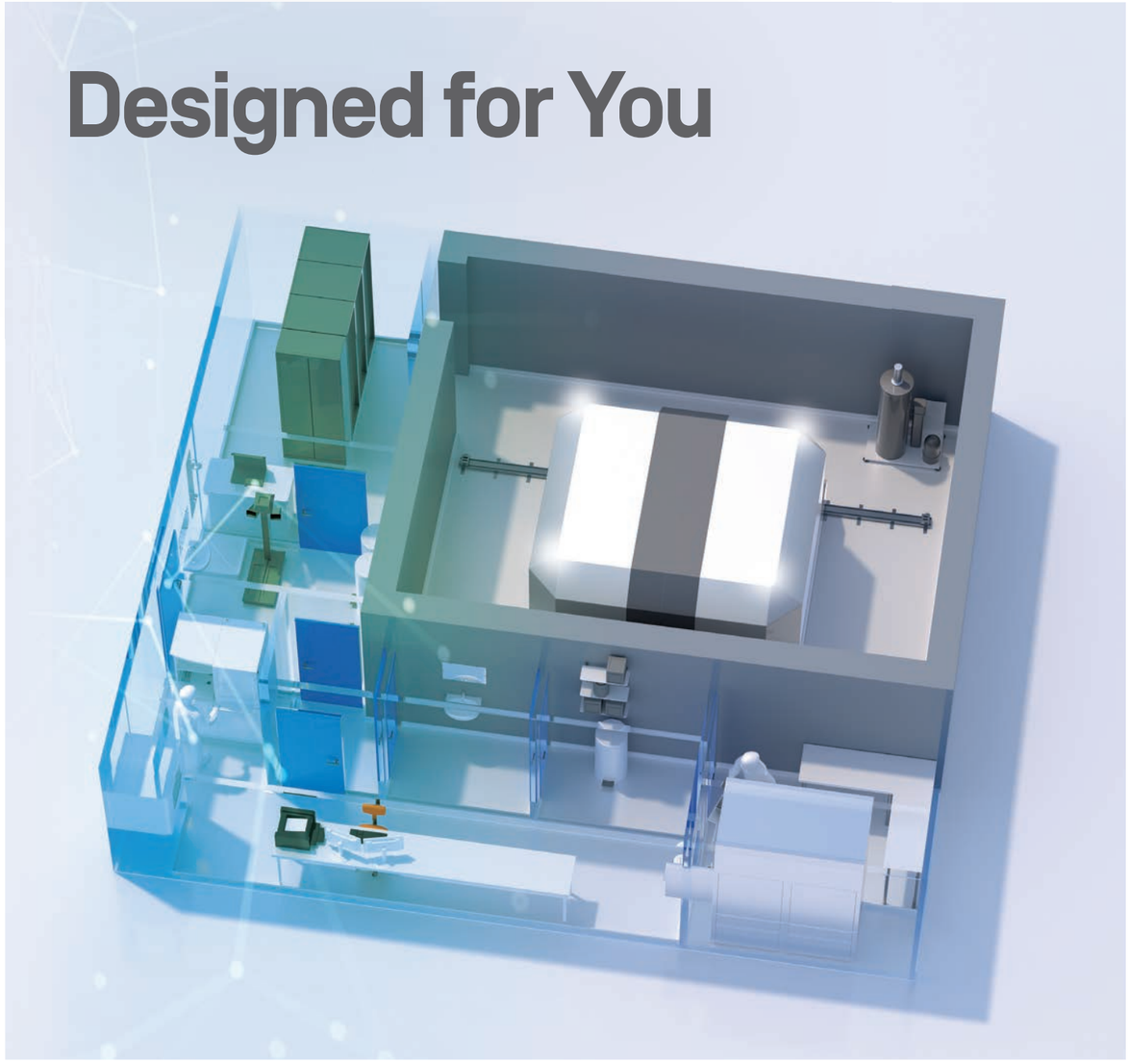
Max current of the Cyclone® KIUBE

HIGHEST PRODUCTION CAPACITY

µA	^{18}F- capacity/run	FDG doses/run[*]
Cyclone® KIUBE 100	2 hours : 10 Ci / 370 GBq	60 - 100
Cyclone® KIUBE 150	2 hours : 16 Ci / 592 GBq	96 - 160
Cyclone® KIUBE 180	2 hours : 20 Ci / 740 GBq	120 - 200
Cyclone® KIUBE 300	2 hours : 30 Ci / 1110 GBq	180 - 300

[*] using Synthera® FDG chemistry [12h shelf life - stability].
May vary according to local condition, transport time and patient schedule.

Designed for You





A SELF-MONITORING CYCLOTRON THAT MAXIMIZES YOUR UPTIME

Unmatched reliability

Cyclone® KIUBE has an unmatched reliability. Its compact design includes redundancy for the main systems (ion sources, strippers, targets, vacuum pumps,...).

The software self-tests prior to production and automatically audits after maintenance, allowing maximized uptime.

The choice of low activation material combined to the optimization of the maintenance tasks ensure the lowest radiation exposure for your staff.

Software in your language

With a control system in several local languages, IBA makes Cyclone® KIUBE globally accessible. It is also very user-friendly, meaning it is easier to train and rotate the staff that operate the system.

The 24/7 helpdesk provides you with peace of mind, with IBA experts analyzing cyclotron data in order to diagnose and solve 95% of problems remotely.





< 100 m^2

1,070 ft^2
PET center area

IntegraLab® ONE

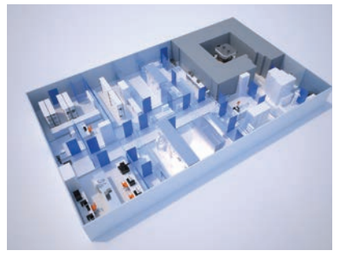
IBA experts designed the **world's most optimized** (c)GMP PET center with a footprint of less than 100m^2 (1,070 square feet) including a 18 MeV self-shielded cyclotron, while allowing a very high production capacity of ^{18}F for FDG and ^{68}Ga.

IntegraLab® ONE is a real ready-to-run integrated Radiopharmacy, designed to ensure a fast, smooth and risk-free set up of your (c)GMP radiopharmacy.

IntegraLab® PLUS

IntegraLab® PLUS is a **fully integrated** solution, combining the equipment and services required to establish a Radiopharmaceutical (c)GMP Production Center tailored to your precise needs.

Our approach is **truly comprehensive**, taking your project from building design to production, with full regulatory compliance, and with the selection, integration, installation and qualification of the equipment needed to meet your radioisotope production goals.



Compact design

Its compact design can fit into any existing cyclotron vault.

Cyclone® KIUBE is also available with a self-shielding option.

Max potential, Max capacity

Nirta® target technology

Complete target range for your extended needs

^{18}F- CONICAL LIQUID TARGET

High power for low enriched water consumption

- Efficient design to achieve 4Ci/ml in 2 hour-run
- Lowest enriched water consumption on the market
- Wide range of volumes available: your production capacity will always match your needs
- Quick and low dose maintenance: less pieces and orings
- Easier Disassembling, assembling and manipulation.



^{18}F

	M	Conical 5	Conical 8	Conical 12	Conical 16
Chemical form			F-		
Target reaction			^{18}O(p,n)^{18}F		
Target material			$H_2{}^{18}O$		
Window material			Havar®		
Insert material			Niobium		
Enriched water volume (ml)	0.7 - 1.0	1.8 - 2.0	2.3 - 2.5	2.7 - 3.0	3.5 - 4.3
Irradiation time (min)	60	120	120	120	120
Recovered activity EOB (Ci) / (GBq)	1.6 / 60	5 / 185	8 / 296	12 / 444	16 / 592



IBA's new conical targets allow us to increase production capacity on a daily basis and reduce enriched water consumption as well.
Design has been simplifed to make maintenance more easy.

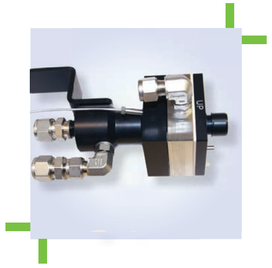
Roel Mooij
Manager R&D Cyclotron and targetry
BV Cyclotron VU, Amsterdam

Max potential, Max capacity

LIQUID TARGETS

Unique solution for ^{68}Ga production

The unique and under patent application (EP 15170854) solution for production of ^{68}Ga for human use has many advantages :
- Seamless automation on the Synthera® platform
- No long-lived ^{68}Ge (271 days) in the final dose
- Production every hour depending on patient schedule
- Simple & cost effective GMP in-house use or for distribution production similar to the ^{18}F process



	^{68}Ga	^{13}N	
Chemical form	$GaCl_3$	NH_3	
Target reaction	^{68}Zn(p,n)^{68}Ga	^{16}O(p,α)^{13}N	
Target material	Zinc salt	H_2O (natural) + 5mMol Ethanol	
Window material	"patent pending"	Havar®	
Insert material	Niobium	Niobium	
Energy degrador	"patent pending"	Graphite	
Beam energy	15 MeV	16 MeV	
Target yield (mCi/µA sat)	11*	38	
irradiation time (min)	60	20	
Insert volume (ml)	3.7	2.3	3.7
Recovered activity EOB (mCi) / (GBq)	120 / 4.5 (upscaling possible)	400 / 14.4	1000 / 37

** depending on target solution concentration*

GAS TARGETS

	^{15}O	^{11}C – CO_2	^{11}C – CH_4
Chemical form	O_2	CO_2	CH_4
Target reaction	^{14}N(p,n)^{15}O	^{14}N(p, α)^{11}C	^{14}N(p, α)^{11}C
Target material	$^{15}N_2$ + 0.5-1% O_2	N_2 + 0.5-1% O_2	N_2 + 5% H_2
Window material	Aluminum	Aluminum	Aluminum
Insert material	Aluminum	Aluminum	Aluminum
Energy degrador	Graphite	na	na
Beam energy	13 MeV	18 MeV	18 MeV
Insert volume (ml)	4 – 5/min (cont. flow)	55	40
Target yield(mCi/µA sat)	30	150	60
irradiation time (min)	online	30	30
Recovered activity EOB (mCi) / (GBq)	80 / 3 (/min cont. flow)	4000 / 148	1000 / 37

SOLID TARGETS

Solid target technology for the production of novel radioisotopes

- Routine production and research programs
- Full packages available including dissolution and purification modules
- Can be installed inside the self-shielding
- Pneumatic transport system between the cyclotron vault and the radiochemistry laboratory.



TYPICAL PRODUCTION WITH SOLID TARGETS – PET/SPECT ISOTOPES

	^{64}Cu[1]	^{89}Zr[1]	^{99m}Tc	^{124}I[1]	^{123}I[1]	^{66}Ga	^{67}Ga	^{111}In	^{86}Y
Half-life	12.7h	78h	6h	4.18d	13.27h	9.49h	78h	67h	14.7h
Target reaction	^{64}Ni(p,n)^{64}Cu	^{89}Y(p,n)^{89}Zr	^{100}Mo(p,2n)^{99m}Tc	^{124}Te(p,n)^{124}I	^{123}Te(p,n)^{123}I	^{66}Zn(p,n)^{66}Ga	^{67}Zn(p,n)^{67}Ga	^{111}Cd(p,n)^{111}In	^{86}Sr(p,n)^{86}Y
Target Material	^{64}Ni	^{89}Y	^{100}Mo	^{124}Te	^{123}Te	^{66}Zn	^{67}Zn	^{111}Cd	^{86}Sr
Energy (on target) (MeV)	14	15	18	15	16	16	16	16	18
Yield (mCi/µAh) [2]	1.34[3]	0.6	95[4]	0.45	4	–	–	–	–

(1) commercial package available
(2) depending on target material thickness
(3) 0.873 mCi/µAh/mg of Ni-64
(4) Theoretical yield

Technical features

	Cyclone® KIUBE 100	Cyclone® KIUBE 150	Cyclone® KIUBE 180	Cyclone® KIUBE 300
HIGH CAPACITY PROTON BEAM				
Energy	18 MeV			
Beam current	100 µA	150 µA	180 µA	300 µA
TARGET FLEXIBILITY				
Number of target ports	8			
Simultaneous extracted beams	2			
Vectio® Beam Transport Line	1 or 4.5 m			
LOW POWER CONSUMPTION				
Stand-by mode	< 3 kW			
Beam-on mode	< 45 kW	< 55 kW	< 60 kW	< 65 kW
COMPACT DESIGN				
Cyclotron weight	18 Tons			
Cyclotron overall dimensions [m]	1.9 x 1.9 x 1.8 [l x w x h]			
Internal room dimensions [m]	3.8 x 3.8 x 2.5 [l x w x h]			
Internal room dimensions with self-shielding [m]	6 x 7 x 3 [l x w x h]		-	-
Self-shielding overall dimensions [m]	5.2 x 4.3 x 2.45 [l x w x h]		-	-



Max potential, Max capacity

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PROTECT +
ENHANCE +
SAVE LIVES

ABOUT IBA

IBA (Ion Beam Applications S.A.), is a cancer diagnostics and treatment company and the worldwide technology leader in the field of proton therapy. The company's expertise lies in the development of next-generation proton therapy technologies and radiopharmaceuticals that provide oncology care providers with premium quality services and equipment, including IBA's leading fully-integrated IntegraLab® system.

ABOUT IBA RADIOPHARMA SOLUTIONS

Based on longstanding expertise, IBA RadioPharma Solutions supports hospitals and radiopharmaceutical distribution centers with their in-house radioisotopes production by providing them global solutions, from project design to the operation of their facility. In addition to high-quality technology production equipment, IBA has developed in-depth experience in setting up GMP radiopharmaceuticals production centers.

ABOUT INTEGRALAB® AND SYNTHERA®+

IntegraLab® is a fully integrated solution combining equipment and services for the development of Radiopharmaceutical Production Centers. IntegraLab® includes the building designed with full regulatory compliance and the selection, integration, supply and installation of suitable high-technology equipment to match your radioisotope production goals.

Synthera®+ is a multi-purpose automated synthesizer for the production of ^{18}FDG, other compounds (^{18}FCH, ^{18}FLT, Na^{18}F, ^{68}Ga peptides ...). This smallest available module on the market is designed to accommodate a wide range of radiochemistry processes.

CONTACT US:

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Tel.: + 32 10 201 275 | Fax: + 32 10 47 59 58 | Info-cyclo@iba-group.com

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